STATISTICS HSSC–I

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Version No.				
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ROLL NUMBER

Answer Sheet No. ____________

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پر کریں۔ **Invigilator Sign.** ____________

Fill the relevant bubble against each question according to curriculum: **Candidate Sign.** ____________

	Question سوال	A	B	C	D	A	B	C	D
1.	Questionnaire method is used in collecting:	True data	Secondary data	Primary data	Published data	○	○	○	○
2.	A graph of cumulative frequency distribution is:	Ogive	Historigram	Pie chart	Frequency polygon	○	○	○	○
3.	The number of classes in a frequency distribution generally should be:	Less than five	Between 5 & 20	Between 10 & 20	More than 20	○	○	○	○
4.	When the classes are 40-44, 45-49, 50-54, then class interval (h) is:	0	1	4	5	○	○	○	○
5.	In a moderately skewed distribution, the mean is 11 and the median is 13 then mode is:	17	15	13	11	○	○	○	○
6.	If $x = 10 + 5u$, $\Sigma fu = -46$, $\Sigma f = 125$, then the arithmetic means is:	11.84	8.16	9.63	10.37	○	○	○	○
7.	Median from the values 9,6,7,4,5,3,7,4 is:	4.5	5	5.5	6	○	○	○	○
8.	Geometric mean for the values 2, -4, and 8 is:	4	-4	-64	Cannot be computed	○	○	○	○
9.	The lack of uniformity or symmetry is called:	Skewness	Dispersion	Kurtosis	Standard deviation	○	○	○	○
10.	If $Q_1 = 25$ *and* $Q_3 = 75$ then quartile deviation is:	50	25	100	12.5	○	○	○	○
11.	If first moment of the number 2 is equal to 5, then mean is:	3	5	7	-7	○	○	○	○

	Question سوال	A	B	C	D	A	B	C	D
12.	Standard deviation is always calculated from:	Median	Mode	Q_1	Mean	○	○	○	○
13.	The independent variable is also known as:	Predictor	Predicted	Explained	Regressand	○	○	○	○
14.	If Laspeyre's price index number=107.22 and Paasche's price index number=105.72 then Fisher's index number is:	106.40	106.47	106.46	106.50	○	○	○	○
15.	If one regression coefficient is greater than one, then other must be:	More than one	Zero	Less than one	One	○	○	○	○
16.	If $\Sigma(X-\overline{X})(Y-\overline{Y})=100$, $r=0.25$, $S_X=8$, $S_Y=5$ then the number of items *i.e.* n is:	40	5	8	10	○	○	○	○
17.	An era of prosperity is an example of:	Cyclical variation	Seasonal variation	Secular trend	Irregular variation	○	○	○	○

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ROLL NUMBER						

STATISTICS HSSC–I

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Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

Q. 2 Attempt any FOURTEEN parts. **(14 x 3 = 42)**

(i) Calculate $[\Sigma(X-4)]^2$, *if* $X = 3, 4, 5, 8$.

(ii) Differentiate between discrete and continuous variable.

(iii) Describe main parts of the table.

(iv) Calculate the class boundaries, and widths for each class.
(a) $(-5)-(-1)$ **(b)** $(-2.75)-1.35$

(v) If the mean of *five* values is 8.2 and *four* of values are 6, 10, 7 and 12 then find the fifth value.

(vi) If sum of 15 values is 300 and by addition of *two* more values, it becomes 360. Find the new values if the ratio between them is 1 : 4.

(vii) The mean of three groups each containing 15 values is 10, 20 and 30. Find mean for all *forty-five* values.

(viii) Write any three properties of arithmetic mean.

(ix) The first *four* moments about mean of a distribution are 0, 4, 6 and 48. Find β_2.

(x) List any three measures of relative dispersion.

(xi) What would be the shape of the distribution if $Mean < Median < Mode$.

(xii) Compute the index number by Fisher Ideal formula from the given data:
$\Sigma p_o q_o = 7800,\ \Sigma p_1 q_o = 11000,\ \Sigma p_1 q_1 = 23000,\ \Sigma p_o q_1 = 16370.$

(xiii) Given $p_o = 5,4,3$ *and* $q_o = 70,75,80$. Find ΣW.

(xiv) If $\overline{X} = 50, \overline{Y} = 110$ *and* $a = 10$. What is the value of 'b' ?

(xv) Find the correlation coefficient from the regression coefficients -0.96 *and* -0.55.

(xvi) Find the slope and Y-intercept of the line whose equation is $4X - 9Y = 18$.

(xvii) Given $r = 0.8, S_{xy} = 20, S_x = 4$. *Find* S_y.

(xviii) Differentiate between signal and noise.

(xix) A series comprises of 60 values each equal to 6. What will be the average and dispersion of the series?

SECTION – C (Marks 26)

Note: Attempt any TWO questions. **(2 x 13 = 26)**

Q. 3 a. The following distribution shows the hourly income of 100 households in a locality: **(06)**

Income	35 - 39	40 - 44	45 - 49	50 - 54	55 - 59	60 - 64	65 - 69
f	13	15	28	17	12	10	5

Calculate the arithmetic mean and show that sum of deviations of values from their mean is zero.

b. Compute mean deviation from mean and median from following frequency distribution. Also calculate the coefficient of mean deviation from mean and median. **(07)**

Daily wages (Rs)	200 - 250	250 - 300	300 - 350	350 - 400	400 - 450
No. of persons	10	20	40	20	10

Q. 4 a. Compute index number of prices from the following data taking 2007 as base using (i) Median (ii) Geometric mean as an average: **(08)**

Commodity	**Average price in Rupees**			
	2007	2008	2009	2010
Tea	24.4	25.8	28.7	28.0
Maize	22.7	24.5	25.5	26.4
Barley	24.6	26.2	27.4	28.1

b. Find regression line Y on X. **(05)**

X	12	14	16	18	20
Y	14	13	12	11	10

Q. 5 a. The following table shows the marks obtained by students in Physics and Statistics. Compute the Correlation Coefficient. **(06)**

Physics (X)	48	40	32	34	30	50	26	50	22	43
Statistics (Y)	76	56	40	50	34	70	56	68	40	57

b. Compute 4-years moving average (centered): **(07)**

Year	2004	2005	2006	2007	2008	2009	2010	2011
Production	81	91	93	84	88	97	101	111